

Auteur: Siebe Mees
Studierichting: Informatica
Oefeningenreeks 4A nr. 6.2

$$\begin{aligned} & \int \frac{-x^2 + x + 1}{x^4} dx \\ &= \int \left(\frac{-x^2}{x^4} + \frac{x}{x^4} + \frac{1}{x^4} \right) dx \\ &= \int \left(-\frac{1}{x^2} + \frac{1}{x^3} + \frac{1}{x^4} \right) dx \\ &= \int (-x^{-2} + x^{-3} + x^{-4}) dx \\ &= \int -x^{-2} dx + \int x^{-3} dx + \int x^{-4} dx \\ &= \frac{1}{x} - \frac{1}{2x^2} - \frac{1}{3x^3} + C \end{aligned}$$

References